

Aaron Kruger

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Professional Experience

University at Buffalo – Buffalo, NY

March 2026 – current

Undergraduate Research Assistant – Design & Fabrication of Large Format 3D Printer

- Used a 6-axis accelerometer and FFTs to analyze resonant vibration frequencies.
- Designed input shaper to negate vibration at resonant frequencies to improve print quality.

Crosby Brownlie – Rochester, NY

May 2025 – current

Engineering Project Management Intern

- Worked alongside engineering and project management teams to plan and execute HVAC&R projects.
- Reviewed drawings, specifications, and submittals to ensure industry standards were met.
- Worked with vendors to ensure delivery of project materials and equipment.
- Managed project documentation such as RFIs, submittals, and change orders through Procore.
- Reported on progress of projects regularly to ensure goals were met in an efficient manner.

Engineering Projects

Reaction Wheel Pendulum

May – July 2025

- Designed a reaction wheel pendulum that uses a DC motor, rotary encoders, and a 3-state controller to balance itself.
- Designed and fabricated using SolidWorks and 3D printer, allowing for rapid prototyping and design tweaks.

Bike Rack Product Design

August – December 2025

- Modeled components of a bike rack and produced appropriate GD&T drawings in accordance with industry standards.
- Conducted static FEA analysis in SolidWorks to improve load-bearing components and find a balance between strength and material cost.

Hydroelectric Dam Design

June – August 2025

- Designed a hydroelectric dam that would span a river in Kenya and generate electricity.
- Conducted necessary fluid mechanics calculations to determine the energy output of the dam and assess viability based on projected payback period.

Model Car Design

August – December 2022

- Designed parts and produced engineering drawings for a model car using SolidWorks.
- Fabricated parts in machine shop using Lathe and CNC Mill
- Made necessary improvements to maximize car's performance across a variety of tests.

Education

SUNY University at Buffalo

Buffalo, NY

- Bachelor of Science in **Aerospace Engineering**
- Dual Major in **Mechanical Engineering**
- Expected Graduation: May 2027
- **Overall GPA: 3.25**
- Dean's List 2025, 2026

Relevant Coursework: MATLAB | Statics | Dynamics | STEM Communications | Product Design | Feedback Controls | Aerodynamics | Solid Mechanics | Materials Science Laboratory | Fluid Mechanics | Linear Algebra | Fluid Mechanics | Heat Transfer | Thermodynamics | Circuit Analysis | Calculus I, II, III | Physics I, II | Differential Equations |

Technical Skills

MATLAB | SolidWorks | Microsoft Office | Arduino | 3D Modeling & Design | Design for Manufacturability | FEA | Python | Fusion360 | ANSYS | PCB Design | Additive Manufacturing | Robotics | GD&T | Risk Analysis | System Control | CFD Analysis |